UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 5: Programming Fundamentals

(CS-308)

Basic Information			
Lab #:	5	**Teacher:**	Arsal Mahmood
Objectives	1. For Loop		
Registration#	2025-ag-9987	**Student Name**	Syed Rameez Haider Rizvi
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : For Loop

1. Calculate the sum of numbers from 1 to 100 and display it on screen by using for loop.

```
#include <iostream> using
namespace std; int main() {
int sum = 0;     for (int i = 1; i
<= 100; i++) {         sum = sum
+ i;
    }
    cout << "The sum of numbers from 1 to 100 is: " << sum << endl;
    return 0;
}
```

```
The sum of numbers from 1 to 100 is: 5050

--------------------------------
Process exited after 0.2087 seconds with return value 0
Press any key to continue . . .
```

2. Calculate sum of even and off numbers separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.

```
#include <iostream> using
namespace std; int main()
{    int lower, higher;
int sumEven = 0;     int
sumOdd = 0;     cout <<
"Enter the lower limit: ";
cin >> lower;     cout <<
"Enter the higher limit: ";
cin >> higher;     for (int i
= lower; i <= higher; i++)
{        if (i % 2 == 0) {
sumEven += i;
        } else {
            sumOdd += i;
        }
    }
    cout << "\n--- Results for range " << lower << " to " << higher << " ---" << endl;
cout << "Sum of Even numbers: " << sumEven << endl;     cout << "Sum of Odd
numbers:  " << sumOdd << endl;
    return 0;
}
```

```
Enter the lower limit: 20
Enter the higher limit: 60

--- Results for range 20 to 60 ---
Sum of Even numbers: 840
Sum of Odd numbers:  800

--------------------------------
Process exited after 3.126 seconds with return value 0
Press any key to continue . . .
```

3. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it by using for loop. Also calculate the total of the grades and the class average.

- >90 = A
- <90 and >80 = B
- <80 and >70 = C
- < 70 and > 60 = D

- <60 = F

```
#include <iostream>
using namespace std; int
main() {
   float marks, totalMarks = 0;
    int studentCount = 10;     for (int i = 1;
i <= studentCount; i++) {
        cout << "Enter marks for student " << i << " (0-100): ";
cin >> marks;
        while (marks < 0 || marks > 100) {
            cout << "Invalid input! Marks must be between 0 and 100. Re-enter: ";
cin >> marks;
        }         char grade;
if (marks >= 90) {
grade = 'A';         } else if
(marks >= 80) {
grade = 'B';         } else if
(marks >= 70) {
grade = 'C';         } else if
(marks >= 60) {
grade = 'D';         } else {
grade = 'F';
        }
        cout << "Student " << i << " Grade: " << grade << "\n" << endl;
totalMarks += marks;
    }
    double average = totalMarks / studentCount;     cout <<
"-----------------------------------" << endl;     cout << "Total
Marks of Class: " << totalMarks << endl;     cout <<
"Class Average:        " << average << endl;     return 0;
}
```

```
Enter marks for student 1 (0-100): 101
Invalid input! Marks must be between 0 and 100. Re-enter: 100
Student 1 Grade: A

Enter marks for student 2 (0-100): 78
Student 2 Grade: C

Enter marks for student 3 (0-100): 67
Student 3 Grade: D

Enter marks for student 4 (0-100): 56
Student 4 Grade: F

Enter marks for student 5 (0-100): 67
Student 5 Grade: D

Enter marks for student 6 (0-100): 78
Student 6 Grade: C

Enter marks for student 7 (0-100): 67
Student 7 Grade: D

Enter marks for student 8 (0-100): 78
Student 8 Grade: C

Enter marks for student 9 (0-100): 67
Student 9 Grade: D

Enter marks for student 10 (0-100): 67
Student 10 Grade: D
```

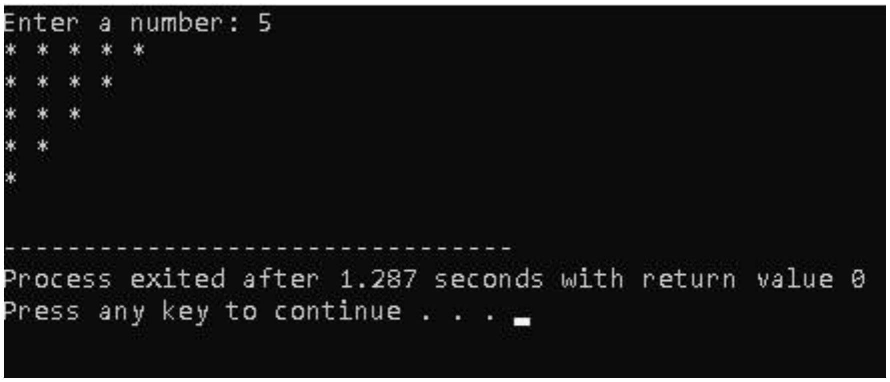
```
Enter marks for student 10 (0-100): 67
Student 10 Grade: D

-----------------------------------
Total Marks of Class: 725
Class Average:        72.5

--------------------------------
```

4. Input a number from user and display a pattern like these on screen by using for loop. For example, user has entered 5, the program should display following pattern on screen.:

(a)
```
*     *     *     *     *
*     * *      *

*     * *

*     *
*
```

```
#include <iostream>

using namespace std;

int main() {

   int n;

   cout << "Enter a number: ";

cin >> n;
```

```
   for (int i = n; i >= 1; i--) {
for (int j = 1; j <= i; j++) {
cout << "* ";
        }
        cout << endl;
   }
return 0;
}
```



P-T-O

(b) 1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

```
#include <iostream>

using namespace std;

int main() {

   int n;

   cout << "Enter a number: ";

   cin >> n;

   for (int i = n; i >= 1; i--) {

       for (int j = 1; j <= i; j++) {
```

```
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

```
Enter a number: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

--------------------------------
Process exited after 3.89 seconds with return value 0
Press any key to continue . . .
```

(c) 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
```

```
            cout << j << " ";
        }
        cout << endl;
    }
    return 0;
}
```

```
Enter a number: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

--------------------------------
Process exited after 2.187 seconds with return value 0
Press any key to continue . . .
```

(d) ? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

? ? ? ? ?

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
cin >> n;     for (int i = 1; i <=
n; i++) {          for (int j = 1; j
```

```
<= n; j++) {            cout << "?

";
```

```
        cout << endl;

    }

return 0;

}
```

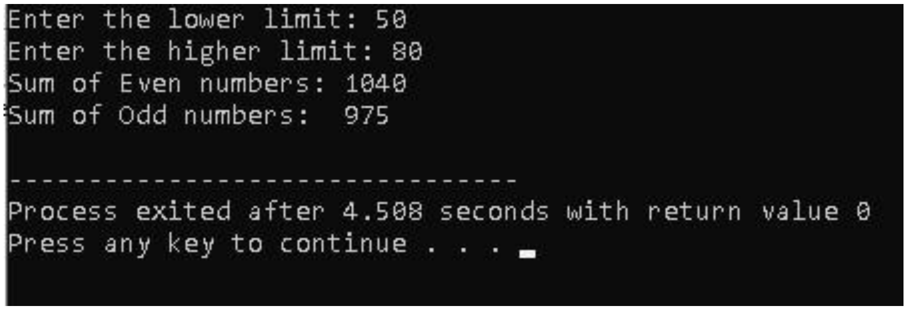
```
Enter a number: 5
? ? ? ? ?
? ? ? ? ?
? ? ? ? ?
? ? ? ? ?
? ? ? ? ?

--------------------------------
Process exited after 1.226 seconds with return value 0
Press any key to continue . . .
```

(Solve all these questions separately. Also, the pattern would change if the user given number is changed)

5. Calculate sum of even and off numbers by using do-while loop separately from a user given range by using for loop. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.

```
#include <iostream> using
namespace std; int main() {     int
lower, higher;     int sumEven = 0;
int sumOdd = 0;     cout << "Enter
the lower limit: ";     cin >> lower;
cout << "Enter the higher limit: ";
cin >> higher;     int current =
lower;     if (lower <= higher) {
        do {            if (current %
2 == 0) {                sumEven
+= current;
            } else {
                sumOdd += current;
            }
current++;
        } while (current <= higher);
    }
    cout << "Sum of Even numbers: " << sumEven << endl;
cout << "Sum of Odd numbers:  " << sumOdd << endl;
```

```
}
    return 0;
 }
```



```
Enter the lower limit: 50
Enter the higher limit: 80
Sum of Even numbers: 1040
Sum of Odd numbers:  975

--------------------------------
Process exited after 4.508 seconds with return value 0
Press any key to continue . . .
```


6. Print the table of user given number from 1 upto user given limit by using for loop. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

```
#include <iostream> using namespace std; int main()
{     int num, limit;     cout << "Enter the number for
the table: ";     cin >> num;     cout << "Enter the
limit: ";     cin >> limit;     cout << "\n--- Table of "
<< num << " ---" << endl;     for (int i = 1; i <= limit;
i++) {         int result = num * i;         cout << num <<
"  * " << i << "  =  ";         if (result < 10 && result
>= 0) {            cout << "0" << result << endl;
       } else {            cout <<
result << endl;            }
return 0;
}
```

```
Enter the number for the table: 3
Enter the limit: 10

--- Table of 3 ---
3  * 1  =  03
3  * 2  =  06
3  * 3  =  09
3  * 4  =  12
3  * 5  =  15
3  * 6  =  18
3  * 7  =  21
3  * 8  =  24
3  * 9  =  27
3  * 10  =  30

--------------------------------
Process exited after 8.927 seconds with return value 0
Press any key to continue . . .
```

7. Write a program to find the factorial of a number using a for loop.

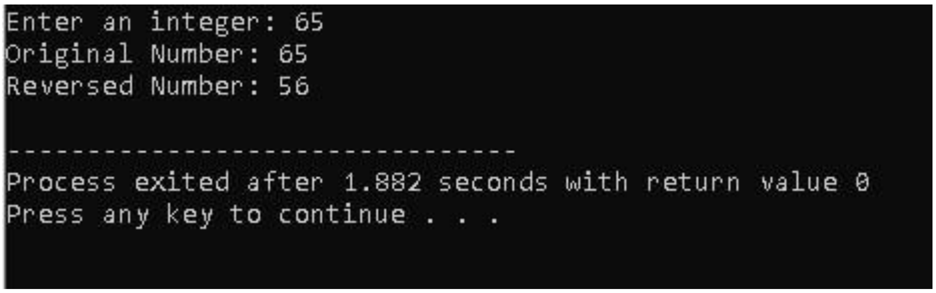
```
#include <iostream>
using namespace std; int
main() {
   int n;    long long factorial = 1;    cout << "Enter a positive integer: ";
cin >> n;    if (n < 0) {        cout << "Error! Factorial of a negative number
doesn't exist." << endl;
   } else {        for (int i = 1; i
<= n; i++) {            factorial *=
i;
       }
       cout << "Factorial of " << n << " = " << factorial << endl;
   }
return 0;
}
```

```
Enter a positive integer: 5
Factorial of 5 = 120

--------------------------------
Process exited after 4.502 seconds with return value 0
Press any key to continue . . .
```

8. Write a program that takes an integer and prints its reverse using a for loop.

```
#include <iostream> using namespace std; int main()
{    int num, reversedNum = 0, remainder;    cout
<< "Enter an integer: ";    cin >> num;    int
originalNum = num;    for (; num != 0; num /= 10) {
```

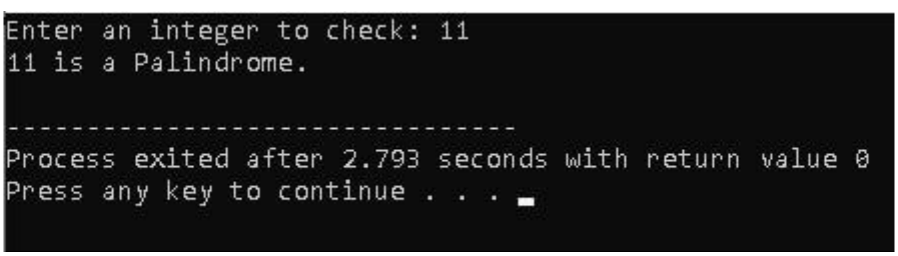
```
    }
remainder = num % 10;            reversedNum =
reversedNum * 10 + remainder;
   }
   cout << "Original Number: " << originalNum << endl;
cout << "Reversed Number: " << reversedNum << endl;
   return 0;
}
```



```
Enter an integer: 65
Original Number: 65
Reversed Number: 56

--------------------------------
Process exited after 1.882 seconds with return value 0
Press any key to continue . . .
```

9. Write a program to check whether a number is a palindrome (same forward and backward) using a for loop.

```
#include <iostream> using namespace std; int main() {
int num, reversedNum = 0, remainder, originalNum;
cout << "Enter an integer to check: ";     cin >> num;
originalNum = num;     for (; num > 0; num /= 10) {
remainder = num % 10;        reversedNum =
(reversedNum * 10) + remainder;     if (originalNum
== reversedNum) {        cout << originalNum << " is
a Palindrome." << endl;
   } else {
       cout << originalNum << " is NOT a Palindrome." << endl;
   }
   return 0;
}
```



```
Enter an integer to check: 11
11 is a Palindrome.

--------------------------------
Process exited after 2.793 seconds with return value 0
Press any key to continue . . .
```

10. Write a program to calculate x^y (x raised to the power y) using a for loop.

```
#include <iostream>
using namespace std; int
main() {
  float x;       int y;       float result =
1.0;     cout << "Enter the base (x): ";
cin >> x;     cout << "Enter the
```

```
exponent (y): ";     cin >> y;     if (y
>= 0) {         for (int i = 1; i <= y;
i++) {             result = result * x;
       }
       cout << x << " raised to the power " << y << " is: " << result << endl;
   }     else {         for (int i = 1; i
<= -y; i++) {             result =
result * x;
       }
       cout << x << " raised to the power " << y << " is: " << 1.0 / result << endl;
   }
return 0;
}
```

```
Enter the base (x): 10
Enter the exponent (y): 3
10 raised to the power 3 is: 1000

--------------------------------
Process exited after 14.35 seconds with return value 0
Press any key to continue . . .
```

11. Print a diamond shape on screen by using for loop.

```
      *
*  *  *
*  *  *  *  *
*  *  *  *  *  *  *
*  *  *  *  *
*  *  *
      *
```

```
#include <iostream>
using namespace std;
int main() {
   int n;
   cout << "Enter the number: ";
cin >> n;     for (int i = 1; i <= n;
i++) {         for (int j = 1; j <= n -
i; j++) {             cout << "  ";
       }
```

```
}
            for (int k = 1; k <= (2 * i - 1); k++) {
    cout << "* ";
            }
            cout << endl;
        }     for (int i = n - 1; i >= 1; i--)
    {         for (int j = 1; j <= n - i;
    j++) {             cout << "  ";
            }
            for (int k = 1; k <= (2 * i - 1); k++) {
    cout << "* ";
            }
            cout << endl;
        }
```

```
    return 0;
}
```

```
Enter the number: 6
          *
        * * *
      * * * * *
    * * * * * * *
  * * * * * * * * *
* * * * * * * * * * *
  * * * * * * * * *
    * * * * * * *
      * * * * *
        * * *
          *

--------------------------------
Process exited after 1.451 seconds with return value 0
Press any key to continue . . .
```

{end of Lab}